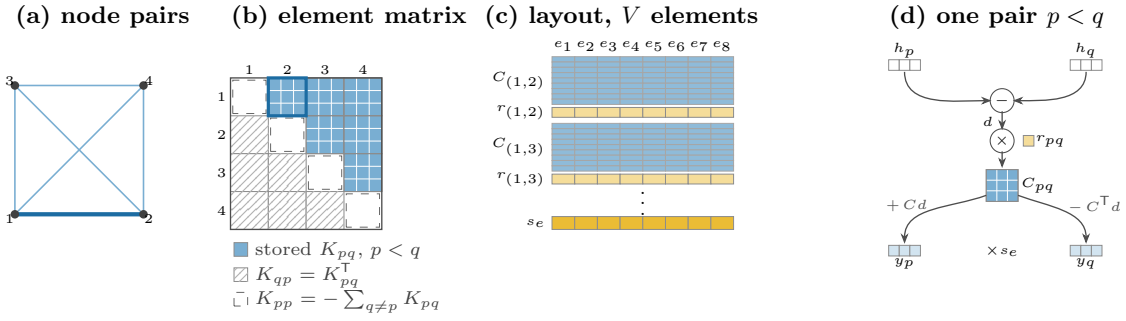

(a) node pairs
(b) element matrix
(c) layout, V elements
(d) one pair $p < q$
1
2
3
4
stored K_{pq}, $p < q$
$K_{qp} = K_{pq}^{\mathsf{T}}$
$K_{pp} = -\sum_{q \neq p} K_{pq}$
e_1 e_2 e_3 e_4 e_5 e_6 e_7 e_8
$C_{(1,2)}$
$r_{(1,2)}$
$C_{(1,3)}$
$r_{(1,3)}$
s_e
h_p
h_q
d
r_{pq}
C_{pq}
$+ Cd$
$- C^{\mathsf{T}}d$
$\times s_e$
y_p
y_q